



Evolution of symbiosis in hermatypic corals: A model of the past, present, and future



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ABSTRACT

This work can be considered a prequel to our previous paper on coral bleaching induced by global warming. We once again investigate, using Finsler geometry, dynamical energy budget theory and nonlinear modular mechanics, the origin of endosymbiosis, between reef-building corals and the algae. We assume their relationship starts out as entosymbiosis, with the algal organism living on the external surfaces of host coral exoskeleton, but with both gradually adapting to each other over evolutionary time-scales. Our main conclusion is that such an evolutionary conversion is possible and indeed is quite likely.

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0. Introduction

Coral reefs have been severely negatively affected by climate change over the past 30 years or more [1–3]. In particular, global warming has caused mass mortalities of zooxanthellate (zooxanthellae-bearing) hermatypic (reef-building) corals on reefs around the world [4,5]. This was first reported to occur in the Eastern Pacific [6,7] and later throughout the world's tropical and sub-tropical seas [8–11]. The cause of coral death in these cases was due to increases in both local and regional seawater temperatures which stress the symbiotic relationship between the coral animal and its endosymbiotic alga, the dinoflagellate *Symbiodinium microadriaticum* [12,13] (commonly called zooxanthellae). This effect (i.e., bleaching) is due to the inability of the zooxanthellae to tolerate increasing seawater temperatures, which in turn causes a breakdown in the symbiotic relationship between the coral animal and the zooxanthellae, and the expulsion of necrotic and

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apoptotic zooxanthellar cells from the holobiont [14,15]. This eviction of zooxanthellae causes “bleaching” of the coral holobiont because pigmentation in hermatypic corals is generally derived from the zooxanthellae, not pigment in the coral tissue. *Symbiodinium* is the weaker of the two partners in terms of temperature tolerance [16–18], and there has been wide-scale loss of zooxanthellae from the corals on a global scale which has caused coral mortality [19–21]. The loss of zooxanthellae causes the tissue of the coral to become transparent, allowing its white calcium carbonate skeleton to be readily viewed through the transparent coral tissue, causing the coral to appear white or “bleached” [22,16].

Zooxanthellae are highly specialized endosymbionts of scleractinian corals and some other reef organisms (e.g., *Cassiopaea xamachana* [23]; *Aiptasia patella*, *Anthozoa*, *Actinaria* [24,25]; and other zooxanthellate symbiotic relationships not unique to the Cnidaria [4]; e.g., *Amphiscolops langerhansii* Platyhelminthes [26], *Tridacna gigas*, *T. maxima*, *T. crocea*, and other *Tridacna* spp Mollusca [27,28]. Most of these are known to bleach under high temperature conditions [29–31]). Endosymbiosis as described by Margulis [32] and Margulis and Fester [33] are interactions taking place within a host and includes intracellular (integration at the cellular level) and/or extracellular (association between the host’s cells). They were, however, initially autonomous free-living organisms similar to *Gymnodinium* sp. and other closely related dinoflagellates. It is believed that early in the evolution of *Symbiodinium*, they were motile, planktonic, and autotrophic [34]. Thus, the change from free-living dinoflagellates to symbiotic zooxanthellae required adjustments within the immune systems of both the anthozoan and the dinoflagellate [35]. We hypothesize that zooxanthellae evolved first from free-living forms, then to entosymbiosis. Heyword and Hichodziejewski (2010) describe entosymbiosis as symbiosis that can occur as (1) “attachment symbiosis” as observed, for instance, when sea anemones ride hermit crab shells and (2) “behavioural symbiosis” where communication is quintessential to making this relationship possible, (e.g. birds cleaning between the teeth of a crocodile) outside a host’s body existing as a permanent/semi-permanent association. Daida [36] describes entosymbiosis as it is possible that the alga started on the surface of a coral but gained a selective advantage by being protected from grazing once inside the anthozoan. This probably ignited the entocommensal relationship. Through time, the complexity of the symbiotic relationship continued to evolve between the coral and the dinoflagellate. The dinoflagellate would have evolved into a sufficiently different organism from its free-living ancestor to become a new and separate genus *Symbiodinium*. The symbiosis became characterized by an obligate inter-dependence between the coral and the zooxanthella, involving an exchange of nitrogenous waste compounds/nutrients and CO₂ from the coral to the endosymbiont, and oxygen and sugars from the alga to the coral.

1. The model basics

We represent this situation with classical ecological equations starting with essentially an arbitrary 2-species system, where species #1 is the host (coral polyp) population of size, N^1 , which contributes nitrogenous wastes to #2, the commensal (zooxanthella), in the amount x^1 , at rate per polyp k_1 , so that

$$k_1 N^1 = \frac{dx^1}{dt}, \quad \frac{dN^1}{dt} = r_1 N^1 - a_1 (N^1)^2 + a_2 (N^2)^2 - a_3 N^1 N^2. \quad (1.1)$$

Likewise, x^2 denotes the amount of photosynthate passed along to #1 (coral) by species #2 (zooxanthella) at rate k_2 , so that

$$k_2 N^2 = \frac{dx^2}{dt}, \quad \frac{dN^2}{dt} = r_2 N^2 - b_1 (N^2)^2 + b_2 (N^1)^2 - b_3 N^1 N^2. \quad (1.2)$$

Here, a ’s and b ’s are all positive and approximately constant (over some pre-assigned time interval of observation) with $(x_1, x_2) \in \mathbb{R}^2$, $(N_1, N_2) \in \mathbb{R}^2$. It is permitted to have the coefficients dependent on the x ’s, provided they are slowly varying. In this case, the coefficients would be approximately constant in some time

interval, possibly one of relatively short duration. The second order ordinary differential equations (SODEs) are also called sprays in the differential geometric literature [37–45].

Let us remark about each of the terms in these two equations.

- Remark 1.1.** (a) If growth rates r_1 and r_2 are approximately equal (denoted r), then this is the pre-symbiont condition necessary for endosymbiosis models and will eventually be invoked as we develop our model [32,46–48]. For now we choose not to implement it.
- (b) $a_1 = r_1/K_a + \beta_1$ and $b_1 = r_2/K_b = \beta_2$ for anthozoan K_a and zoox K_b , the carrying capacities. The zooxanthellae are considered to have quite a large carrying capacity in an interval of time during settlement onto an external surface of a coral host. This is the situation where b_1 is quite small because K_b is large. There is, however, a stable steady-state for this model and it is given below in (1.7). The negative feedback constant, K_a , for logistic growth of the host, is retained from its independent lifestyle of previous epochs. This is not the case for the zooxanthellae. Once it breaks through the hosts immune system barrier and begins giving and receiving nutrition, it will automatically be constrained by an upper limit, which is K_b .
- (c) a_2 is small, since for the entosymbiosis interaction, the host, #1, gains little or nothing from the commensal, #2.
- (d) Negative signs for mixed terms, $a_2 N^1 N^2$ and $b_2 N^1 N^2$, indicate competition for resources for interactions schemes involving entosymbiosis.
- (e) $a_1 = r_1/K_a + \beta_1$, $a_3 = \beta_2 = b_1$, $b_3 = 2r_1/K_a + \beta_1$.

We see from (e), above, that the coral’s self-inhibition, a_1 , is less than the negative effect of competition, b_3 , on the commensal zooxanthellae, while the effect on the coral host, a_3 , is a relatively small perturbation. Yet, the overall benefit to the zooxanthellae population must outweigh the negative impact it receives while residing externally on the host surface. This is characteristic of entosymbiosis.

Theorem 1.2. *Production given by (1.1) and (1.2) preserves $F(x, dx)$ along any solution γ , while total energy, $\int_\gamma F^2(x, \dot{x}) dS$ achieves a minimum if and only if $\beta_1 = \beta_2 = 0$.*

This F is referred to as *formal cost*, whereas its square is *Medawar’s growth energy*, (MGE) [49].

Proof. The associated 2nd order system of equations, obtained by combining (1.1) and (1.2) with conditions 1 through 5 and $\beta_1 = \beta_2 = 0$, are precisely the Euler–Lagrange equations obtained through fixed end-point variation of the total energy, where F^2 is

$$F^2 \left(x, \frac{dx}{dt} \right) = (N^2)^2 \cdot e^{2N^1/N_2 + 2\sigma(x)}, \quad (1.3)$$

which is defined on $T\Omega$, the tangent bundle with deleted origin [38,39]. Furthermore, using parameter S (size or total production) with

$$e^{rt} \cdot dt = dS = F(x, dx) \quad (1.4)$$

and the linear function

$$\sigma(x) = (a_1 - b_3)x^1 + a_1x^2, \quad (1.5)$$

it follows that the preservation condition,

$$\frac{dF}{dS} = 0, \quad (1.6)$$

hold true along any solution γ of (1.1) and (1.2). This is a minimum in the case of vanishing β ’s, because then, and only then, do the Euler–Lagrange equations for F^2 become identical to (1.1) and (1.2). $\square$

- Remark 1.3.** (a) In the evolutionary past, a_2 , b_2 , a_3 , and b_3 , in (1.1) and (1.2), would have been close to vanishing. The two species would then have been free-living, independent, motile, planktonic, and, to some extent, autotrophic. This is described by logistic growth for each component.
- (b) As time progressed, entosymbiosis would have arisen enabling the zooxanthellae to secure protection from the external marine world by residing on an exposed anthozoan surface outside anthozoan tissues. In order for this to have happened, the immune system of each separate component would have had to accommodate the presence of the other, recognizing it as “self”, at least to some extent. So, coefficients a_3 and a_2 would be close to vanishing. In this way, a commensal relationship, mediated by exchange of chemical compounds produced by each component, could arise. The unique steady-state is stable and given, assuming $b_1 \neq 0$, as

$$N_*^1 = \frac{r^1}{a^1}, \quad N_*^2 = \frac{a_1 r_2 - b_3 r_1 + \sqrt{(a_1 r_2 - b_3 r_1)^2 + 4b_1 b_2 r_1^2}}{2a_1 b_1}. \quad (1.7)$$

If $b_1 = 0$, the stable steady-state is modified to

$$N_*^2 = \frac{r_1^2 b_2}{a_1(r_1 b_3 - r_2 a_1)}. \quad (1.8)$$

Note that the denominator of (1.8) must be positive. For consistency, it follows that the first term in (1.7), inside the braces, is negative while the radical term is larger, so zooxanthellae density is positive.

2. Evolution via Heterochrony

Now, consider the case where β 's do not vanish. In order to describe this consider the asymmetric Wagner connection [38,50,49].

$$\Gamma_{jk}^i = G_{jk}^i + \delta_j^i \partial_k \tilde{\sigma}, \quad (2.1)$$

$$\tilde{\sigma}(x) = \sigma(x) + \beta_1 x^1 + \beta_2 x^2, \quad (2.2)$$

where $G_{ii}^i = a_i$, and the other four of G_{jk}^i vanish (there are 8 possibilities for this symbol, unchanged when the lower indices have reversed order). Note, $T_{jk}^i = \Gamma_{jk}^i - \Gamma_{kj}^i \neq 0$, unless $\beta_1 = \beta_2 = 0$.

Let us now recall the model of evolution, called Heterochrony, or time-sequencing change, of developmental processes within the colonial entosymbiont [49–53,37]. What characterizes heterochrony is the changes in the array of interaction coefficients, induced from a change of size (or joint production parameter) S , to a new one, P , according to the line-integral formula,

$$\frac{dP}{dS} = C \cdot \exp \int_{\gamma} (\tilde{\sigma}(x(S)) - \sigma(x(S))) dS \quad (2.3)$$

where γ is a solution $\{x^i(S)\}$ to (1.1) and (1.2), completely determined by $x^i(0)$ and $N^i(0)$ and C is an arbitrary constant (set equal to unity for convenience). The production cost functional, F^2 , in (1.3) and (1.4)–(1.6) must now be modified to

$$\tilde{F}^2(x, N) = e^{2\tilde{\sigma}(x) - 2\sigma(x)} F^2(x, N). \quad (2.4)$$

Considering (1.1) and (1.2) with conditions of Remark 1.1 invoked, it follows by direct calculation that along any solution γ , the total derivative with respect to the new size parameter, P , satisfies

$$\frac{d\tilde{F}}{dP} = 0, \quad (2.5)$$

which completes the proof of

Theorem 2.1. *Evolution by heterochrony enables the entosymbiont to conserve both MGE and the formal cost relative to the production parameter, P . That is, $d\tilde{F}/dP = 0$, along any solution trajectory γ [49,50] [54, appendix].*

Remark 2.2. (a) The time-change process is applied after the change from clock time, t , to size parameter, S , in (1.4). Then the heterochronic change from S to the new production parameter, P , is executed. This is followed by return to clock time t , via (1.4), but with S replaced by P , and F , replaced by $\tilde{F}$. As a consequence, the original ecological interaction pattern $G_j k^i$, is augmented with the linear sum ${}_i x^i$, as in (2.2), and becomes Γ_{jk}^i , a new asymmetric ecological interaction pattern. If we augment (1.5) with the negative sum of quadratics, $\sigma_i (x^i)^2$, then production is Jacobi stable, since the stability index, R (i.e., scalar curvature), is

$$R = 2\sigma_2 e^{-2\rho(x)-2N^1/N^2} \quad (2.6)$$

$$\rho(x) = \sigma(x) - \sigma_1 (x^1)^2 - \sigma_2 (x^2)^2 - \frac{1}{2} v_3 x^1 x^2. \quad (2.7)$$

Note that since x_i are slowly changing, the coefficients G_{jk}^i , now linearly dependent on x_i , will be approximately constant. Yet, when $\sigma_2 > 0$, R is positive, so production is Jacobi stable, a concept concerned with the longer evolutionary time-scale, S , rather than the shorter ecological encompassing one to several generations, t . Note that R vanishes if and only if $\sigma_2 = 0$.

- (b) Interestingly, x_1 and $\nu_3 x^1$ enter the interaction coefficients, G_{jk}^i , of the geodesics, via (2.7), but have no effect on R , the Jacobi stability index. We conclude that it is what the zooxanthellae produce, not what the anthozoans produce, which ensures stability of joint production in the entosymbiont, although each has an effect on the interaction coefficients.
- (c) In Analytical Modular Dynamics (AMD) the net amount of protein produced is assumed to be an allometric surrogate of total biomass, for each of the two types of producer. In the model equations, the coefficients depend on x_1 and x_2 and represent efficient production/reproduction. Nevertheless, it is assumed the net accumulation is slow enough in some time interval, so the interaction coefficients, Γ_{jk}^i , are approximately constant.
- (d) In order to model evolutionary progression from independent lifestyles to entosymbiosis, the positive coefficients, a_2 , b_2 , a_3 , and b_3 , get larger over time. Also, production must become robust enough to withstand random perturbations in the ambient environment. These requirements hold true and are consequences of the mathematics. Moreover, from a mathematical perspective, to maintain a commensal life style the competitive coefficient, a_2 would have had to eventually approximate the negative feedback term, b_1 . This is a constraint required by the theory for quadratic 2-dimensional Finsler theory which we here apply [55, appendix].

As a matter of fact, once entosymbiosis has been integrated into the genes of the zooxanthellae and stabilized, it would be impossible for joint production to allow both anthozoans and zooxanthellae to be on equal footing. It is this equality that was a key point in our published formulation of the endosymbiotical model of bleaching [56–58]. The present, ento-model and the published endo-model, are very different geometrically speaking. It is impossible, even over long time periods, for either one to transform into the other. The first reason is for taxonomic reasons and lack of relatedness. The second is a mathematical reason, namely, the interaction patterns must involve the production variables, if the system is to be stable. However, if the role of the products (x 's) in the interactions is minimized, say, at a stage where zoox presence is relatively small, that is, N^1 , is close to zero, as could happen

in the beginning of their association. It is now apparent that securing a stable entosymbiosis would be an evolutionary deadend. There would not be enough independence for the zooxanthellae to allow the known large numbers of clades to come into existence through natural selection [56,59].

Considering an evolutionary progression towards increasing integration in the entosymbiont, the reproductive processes of anthozoa must be supported by zooxanthellae, and this must be reciprocal. To represent reproduction, a positive term of the form, $a_3(N^2)^2$ must appear in the anthozoa equation, (1.1), and likewise, in the zooxanthellae equation, (1.2). Such would have put the two components on equal ecological footing culminating in characteristic holobionts that are the modern scleractinian corals. But, does this follow from the present model? That is, can we prove that the ento-model, herein described, evolves (via heterochrony) into the endomodel in [56]? We already know the answer. It is: no.

- (e) At this juncture, the model shows that with conditions in Remark 1.1 invoked, the total (joint) production process (1.1)–(1.5), (1.6), (2.1)–(2.5) is conservative, but not optimally so, because the production curves are not geodesics. Rather, they are autoparallels of the Wagner connection, (2.1), (2.2). It is this Wagner property that ensures the conservation of energy in terms of total production, P , as in (2.5). The energy functional for the endo-model [56] is given by

$$F = \sqrt{(N^1)^2 + (N^2)^2} \cdot \exp \left[(p^2 + 1)\alpha(x) + p \cdot \arctan \left(\frac{N^1}{N^2} \right) \right] \quad (2.8)$$

with

$$\alpha(x) = \alpha_1 x^1 + \alpha_2 x^2 - \frac{1}{2} (\sigma_1 (x^1)^2 + \sigma_2 (x^2)^2 + \nu x^1 x^2),$$

while that for the ento-model above, using (2.7), is

$$L = N^2 \cdot e^{\rho(x) + N^1/N^2}. \quad (2.9)$$

The corresponding curvatures are, respectively,

$$R = (p^2 + 1)(\sigma_1 + \sigma_2) \cdot \exp \left[-2\alpha(x) - 2p \cdot \arctan \left(\frac{N^1}{N^2} \right) \right] \quad (2.10)$$

and

$$R = 2\sigma_2 e^{-2\rho(x) - 2N^1/N^2}. \quad (2.11)$$

When N^1 (always positive) is very small, $\arctan(N^1/N^2)$, is well-approximated by N^1/N^2 . So, if we choose $p = 1$ and $\sigma_1 \sim 0$, it is clear that F well-approximates L . It is therefore proved that for a small zooxanthellae population, and a normalized choice of Finsler parameter p , a transformation of L into F is a possible evolutionary passageway. It allows escape from the lopsidedness of entosymbiosis into the laissez-faire world of equal-rights for partners in an endosymbiotic association.

It should be mentioned that 2-dimensional Finsler geometries are characterized by a pair of invariants, namely, R , the scalar curvature, and $\mathcal{I}$, the principal scalar [54]. Two Finsler geometries on an open set Σ in x -space are isometrically isomorphic if and only if R and $\mathcal{I}$ are identical relative to a fixed coordinate system in Σ . The endo-model has $\mathcal{I}^2 = 4p^2/(p^2 + 1)$, while the ento-model has $\mathcal{I}^2 = 4$. Obviously, no matter how large p gets, these invariants can never be exactly the same. Yet, it is noteworthy that for a p -value as small as $p = 20$, one finds $\mathcal{I}^2 = 3.9900$. It is because of these mathematical reasons that the present entosymbiosis model is allowed to transform into the endosymbiosis model [56].

In the event that entosymbiosis has not become stable, the interaction coefficients must be approximately constant. Here, we may invoke mentally, what Stephen J. Gould referred to as ‘putting the gears of Needham’s engine into neutral’, thereby taking advantage of the disassociation, during some phases of ontogeny, of the different systems (“gears”) making up the developing organism (“engine”) [61]. We mean, of course, Sir Joseph Needham, the early 20th century biologist who studied development [60]. We have made a mathematical model of his “gear shifting”, which we call the “Finsler gate”, as it involves the 3 different Finsler metrics (dimension 2) with constant coefficient geodesics, each regarded as a “path leading away from the (Finsler) gate”. The Finsler gate is the single equivalence class consisting of the three interaction patterns, each of which can be transformed into either of the other two, by a nonsingular coordinate transformation (hence, the equivalence). When the engine of ontogeny is in neutral, there is a choice to be made between the 3 possible paths lying in the ‘epigenetic landscape’ of development, using imagery of C.H. Waddington. We may imagine 3 valleys spreading out on this landscape. Each has a geometry distinct from the other two and each leads away from the gate. As a rolling embryo moves along its chosen valley, the valley gets deeper and more details of its shape become defined. This expresses, in our coral/algae case, the increasing integration of the anthozoan and zooxanthellar components. When x ’s are small, there is not much integration in the valley, but as time grows longer, the role of the x ’s can become more important and would be revealed in the curvature R , with different Finsler metrics causing the valley paths to behave differently, some Jacobi stable, some not. We now construct a table to indicate just what products (x ’s) cause which changes to the symbiont as reflected in the curvature, R .

There are more choices which can be made at the Finsler gate allowing evolution to remold the epigenetic landscape via phenotypic plasticity in the genome generated.

3. Finsler gate: The plastic deformation of phenotype

This section concerns the unique set of 8 constant-coefficient *sprays*, 3 of them, G^I , G^{II} , and G^{III} , are geodesics, but all 8 of them satisfy conservation laws. Moreover, the 3 geodesic sprays forming the Finsler Gate are termed equivalent, but are also called elastic deformations. Either of these 3 is elastically deformable (reversible coordinate changes) into each of the other two. The remaining 5 are plastic deformations of these three. The coordinate transformations describing the plastic deformations are not reversible and are technically, non-integrable. They model phenotypic plasticity.

- (1) G^I , G^{II} , G^{III} denote the three constant coefficient geodesic sprays. They are transformable, any one into any other, via an appropriate coordinate transformation (non-singular). They are by definition elastically deformable into each other and in this sense are equivalent. This means there are reversible transformations converting any of the three interaction patterns (G ’s) into either of the remaining two. The equivalence class of these three geodesics defines the Finsler Gate. The geodesics equations exhibiting the three interaction patterns are given below.
- (2) These 8 coefficient arrays, ${}^{(\alpha)}\Gamma_{jk}^i$, are time-sequencing equivalent to each other as long as non-singular coordinate transformations are permitted. Technically, they are projectively equivalent to straight-line geodesics of Cartesian space. Exactly 5 are obtained by non-reversible coordinate transformations of G^I , G^{II} , and G^{III} , the so-called plastic deformations. Together, these 8 interaction schemes form a “Garden of Forking Paths” from Jorge Luis Borges’ “Labyrinths”, where each path leads away from the Finsler Gate into the epigenetic landscape of C.H. Waddington.
- (3) The concept of heterochrony must involve an external influence expressed as a vector field, C^i , to be understood as the external “driving force” behind the time-sequencing change. But note that time-sequencing equivalence is not heterochronic equivalence. The elastic deformations (reversible coordinate changes) are separate from time-sequencing changes. But projective equivalence is time-sequencing

equivalence plus any elastic coordinate changes. Projective change is a concept which provides a mathematical model of heterochrony. There are two parts to it. In addition to the actual time change ($s \rightarrow p$) along a production trajectory, an (elastic) transformation of substances produced can occur. The reversible coordinate change models this conversion which takes place during ontogeny. This is not a plastic transformation.

Setting $r_j^i = \lambda \delta_j^i$, $\lambda > 0$, we introduce the total joint production parameter, s , and use $dp = \frac{1}{\lambda} e^{\lambda t} dt$ to change (1), (2) into the spray

$$\frac{d^2 x^i}{dp^2} + {}^{(\alpha)}\Gamma_{jk}^i \frac{dx^j}{dp} \frac{dx^k}{dp} = 0. \quad (3.1)$$

Theorem 3.1 (*The Finsler Gate*).

- (a) There are exactly 3 constant coefficient arrays G^I , G^{II} , G^{III} , exhibited below (defining the Finsler Gate), with (3.1) having solutions $\gamma(s)$, which are geodesics of one of the 3 conformally Minkowski Finsler metrics, i.e., $e^{\varphi_1} F^I(y)$, $e^{\varphi_2} F^{II}(y)$, $e^{\varphi_3} F^{III}(y)$, where $\varphi_1, \varphi_2, \varphi_3$ are certain linear functions of x^1, x^2 (adapted coordinate charts). There are no other 2-dimensional Finsler metrics yielding geodesics. The three cost functionals are as follows:
- (i) $F^I = e^{[(1+\beta) \ln y^2 - \ln y^1]/\beta}$, constant $\beta > 0$, $\mathcal{I}^2 = \frac{(\beta+2)^2}{\beta+1} > 4$,
 - (ii) $F^{II} = |y^2| e^{y^1/y^2}$, $\mathcal{I}^2 = 4$.
 - (iii) $F^{III} = \sqrt{(y^1)^2 + (y^2)^2} e^{L \arctan(y^1/y^2)}$, constant $L > 0$, $\mathcal{I}^2 = \frac{4L^2}{1+L^2} < 4$.
- (b) There are 5 constant coefficient arrays ${}^{(1)}\Gamma_{jk}^i, \dots, {}^{(5)}\Gamma_{jk}^i$ each giving rise to a conservation law. There are uniquely associated 5 Finsler cost functionals of the form $e^{\sigma(x)} F(y)$, each constant along solutions $\gamma(p)$ of its corresponding spray, (3.1) above. That is,

$$\frac{d}{dp} \left(e^{\varphi(\alpha)} \cdot {}^{(\alpha)}F \right) = 0$$

along $\gamma(s)$. These 5 are not geodesics, but they do conserve cost and energy.

The proofs may be found in the appendix of [54,50].

Each of the coefficient arrays forming the Finsler Gate, GF, yields geodesics each of (i), (ii), (iii). These, in turn, are classified, up to isometry, by 2 invariants: (1) curvature condition, $K = 0$, and (2) the numerical value of $\mathcal{I}$, the *principle scalar*, which varies through all the positive reals, R^+ , splitting this set into 2 disjoint open intervals. Note that (iii) is the only cost functional that allows the Euclidean metric case, $L = 0$. Moreover, the classical ecological interactions of competition, parasitism, and mutualism can only be derived from (i) via certain plastic deformations [52].

Let us now display the 3 constant (positive) coefficient geodesics that define the Finsler Gate. For each case we adjoin the production equations $dx^i/dp = y^i$, $i \in \{1, 2\}$:

$$G^I : \frac{dy^1}{dp} + \beta Q_1 (y^1)^2 = 0, \quad \frac{dy^2}{dp} + \beta Q_2 (y^2)^2 = 0; \quad (3.2)$$

$$G^{II} : \frac{dy^1}{dp} + a_1 (y^1)^2 = 0, \quad \frac{dy^2}{dp} - b_2 (y^1)^2 + 2a_1 y_1 y_2 = 0; \quad (3.3)$$

$$G^{III} : \begin{cases} dy^1/dp + 2Q_2 y^1 y^2 + Q_1 [(y^1)^2 - (y^2)^2] = 0, \\ dy^2/dp + 2Q_1 y^1 y^2 + Q_2 [(y^2)^2 - (y^1)^2] = 0. \end{cases} \quad (3.4)$$

The plastic deformation equations arising from (3.2) and (3.3) are respectively

$$\begin{aligned}\frac{dy^1}{dp} &= -[\beta Q_1 + \sigma_1](y^1)^2 - \sigma_2 y^1 y^2, \\ \frac{dy^1}{ds} &= -[\beta Q_2 + \sigma_2](y^2)^2 - \sigma_1 y^1 y^2,\end{aligned}\tag{3.5}$$

where $\beta, \sigma_1, \sigma_2$ are positive constants, and $\varphi_1 = (\sigma_1 - Q_1)x^1 + [\sigma_2 + (1 + \beta)Q_2]x^2$;

$$\begin{aligned}\frac{dy^1}{dp} &= -(a_1 + \beta_1)(y^1)^2 - \beta_2 y^1 y^2, \\ \frac{dy^2}{dp} &= -\beta_2 (y^2)^2 + b_2 (y^1)^2 - (2a_1 + \beta_1)y^1 y^2,\end{aligned}\tag{3.6}$$

with constants as in (1.1), (1.2), and $\varphi_2 = (a_1 + \beta_1 - b_2)x^1 + (a_1 + \beta_2)x^2$. Also,

$$\frac{dp}{ds} = A \cdot \exp\left(-\int_{\gamma} R(x)ds\right) \quad \text{and} \quad R(x) = \sigma_i x^i,$$

$\gamma(s)$ being a solution trajectory in x -space.

If we assume that there was one ancestral species of *Symbiodinium* early in the evolution of zooxanthellae, then there must have been substantial adaptive radiation since that time, in order for us to observe the surviving eight to nine clades that exist today. Now, in the recent, we are left with these 8–9 clades, and an associated 40–50 sub-clades, each of which must be expected to have some special adaptation. It is possible that at least some of these sub-clades are differentiated with respect to temperature tolerance. Once the temperature limits of these clades and sub-clades have been exceeded, however, the population of corals hosting them will likely die or be forced to find a new symbiont with a broader temperature tolerance range [62].

Firstly, it is important to understand that most species that have ever existed on Earth are now extinct. It is likely that this is also the case with the corals and dinoflagellates, including zooxanthellar clades and sub-clades.

There have been numerous changes in the Earth's environment through evolutionary time. These have included climatic changes such as global warming and cooling. These in turn have affected the evolution and phylogenetic radiation of the coral, the endosymbiotic algae, and the multiple clades in *Symbiodinium*, the remnants of which we now observe.

4. Additional background

Scleractinian corals and their endosymbionts were derived from two very different phylogenetic origins, not the least being that one is an animal and the other a plant. Scleractinian corals as we know them today became recognizable as a taxonomic group during the Middle Triassic, some 200–250 years ago [63]. On the other hand, *Symbiodinium* spp., the endosymbiotic algae which occur inside the corals in intercellular spaces, is a dinoflagellate and is closely related morphologically to *Gymnodinium* spp. *Symbiodinium* spp. or its ancestor(s) probably also arose multiple times during the Tertiary, 2.6–65.5 MYA. The symbiosis permitted high rates of calcification of the coral's skeleton to occur, and that emergence is well preserved in the fossil record. The symbiosis which evolved was characterized by an obligate inter-dependence between the coral and the zooxanthellae.

In order for symbiosis to evolve between these two disparate taxa, a number of critical criteria must have been met, and each of these criteria may require a mutation or set of mutations in both organisms [62,64].

- The first involved the immune system of the coral. Corals are able to discriminate between proteins of various organisms, including other corals, as “self” and “non-self” [65]. The coral would have to perceive the foreign “invader” as “self” and not attack the cell in order for the invasion to be successful.
- Secondly, the coral would have to be perceived as “self” by the alga, which would help to insure successful penetration of the dinoflagellate into the tissue of the coral. Since this is a protein recognition system [66], proteins would have to be configured in such a way on both sides to allow this to happen.
- Third, the dinoflagellate will have entered a new environment—the mucus-like, inter-cellular environment of the coral. The dinoflagellate must be adapted to survive, grow, and reproduce in this new environment.
- Fourth, asexual reproduction would probably be favored in the dinoflagellate while living inside the coral. The reason for this is that if immuno-compatibility had been reached between the coral and the dinoflagellate, stabilization of that relationship would be adaptively advantageous; thus, the elimination of genetic recombination through sexual reproduction would function best in this new symbiotic relationship. In addition, the zooxanthellae would not have to locate a mate and then translocate through a gel environment to reach it.
- Fifth, because of the new environment of the dinoflagellate, it would have to obtain its metabolic needs from that environment—the coral tissue. Thus, CO₂ and nitrogenous nutrients necessary to maintain survival, growth, and reproduction would have to be derived from the coral, since the dinoflagellate would no longer be in direct contact with seawater. Thus, the host coral would be the most likely place to obtain these.
- Sixth, the coral in its primitive (non-symbiotic state) would be obtaining its O₂ and nutrients (sugars, proteins, etc.) directly from the seawater and the plankton it ingests. In the first instance of invasion by the dinoflagellates, it would have no need for the O₂ and the carbohydrates/sugars produced by them. However, with time, these metabolic needs may well have been more readily met by accepting them from the dinoflagellate than from the external environment.
- Seventh, the presence of the dinoflagellate within the coral assisted and facilitated the precipitation of CaCO₃ in the coral, permitting it to grow faster, secrete a stronger skeleton, and outcompete its neighbors for space more easily.

This symbiotic relationship has been successfully adopted across a number of phyla, including the Mollusca (*Tridacna* spp. [27,28]), Platyhelminthes (flatworms [26]), and other cnidarians.

There are examples, primarily terrestrial, whereby symbionts have been exchanged in symbiotic systems—even where the new symbiotic partner is phylogenetically distant [67,68]. For example, ectomycorrhizal fungi have replaced arbuscular mycorrhizal fungi in the roots of trees [69]. Yeasts have replaced *Buchnera* bacteria in aphids [70]. With respect to dinoflagellates, endosymbiotic algae have replaced plastids that contain peridinin [71]. The capability of corals to accept other symbionts is not yet known and has not yet been investigated, but can be modeled based on current knowledge.

The question now arises, can the host accept a new potential symbiotic partner, and can it do so in time to keep pace with global warming rates? And what taxa would serve as the potential replacement species? Belda-Baillie et al. [72] have hypothesized that the probability of many characters aligning – in both the host and symbiont – to accommodate a change in symbionts is negligible. We hypothesize, however, that even low probabilities of positively adaptive mutations occurring are worthy of consideration to understand potential directions in the future evolution of scleractinian corals.

There are several possibilities which exist regarding alternate symbiotic relationships:

- *Prochloron* is a cyanobacterium which is symbiotic in some sponges and didemnid ascidians [73], such as *Didemnum molle*. It can be a free-living organism and can also be symbiotic.

- Some other species of phytoplankton or algae are also known to serve as symbionts in some invertebrates, such as *Chlorella* [74,75].
- Another possibility is marine bacteria, either free-living or currently living in association with coral. Corals are now known to possess a microbial community, including marine bacteria, in their surface mucus layer [76].

In order for a new symbiotic relationship to develop, all of the criteria discussed above regarding the coral accepting the new potential symbiont would also have to be met before it could be accepted. The probability of success of a new symbiotic relationship occurring would vary greatly between taxa. For example,

- A photosynthetic alga or bacterium could be adopted by the coral. It is possible, however, for the photosynthetic organelles (i.e., chloroplasts) or structures to be extracted from a exploited target organism, leaving a fully photosynthetic host with no associated living endosymbiont. There are precedents for such.
- To be more specific, the possibility exists that the coral may ingest the dinoflagellates or another phytoplankton or alga, digest or pierce the cell wall during the digestive process, ingest the cytoplasm along with the intact chloroplasts, and move the chloroplasts to the intercellular areas to replace the zooxanthellae [77]. In this case, the temperature tolerance of the symbiont would become moot; only the temperature tolerance of the host and the chloroplasts would be important.
- The relative probability of a photosynthetic bacterium associated with the mucus on the exterior of the coral being brought into the interior and then becoming symbiotic is relatively low. Such a relationship would require immune system adaptations to support endosymbiosis.
- It is also possible that there could be evolutionary movement away from symbiotic algae back to heterotrophy, which is descriptive of azooxanthellate corals.
- It is important to note that any new endosymbiont or photosynthetic machinery must possess thermal characteristics capable of tolerating seawater temperatures in the current environment—an environment which is causing the loss of zooxanthellae. Different symbiotic paths may be adopted by different populations of corals.

The survival of scleractinian corals in the projected hyper-tropical zone of the Earth [62] is dependent upon one or more of these sets of adaptations to occur prior to temperatures surpassing predicted critical levels. We predict that many coral species will survive by their populations expanding into higher latitudes, following the concomitant movement of their climatic zones to higher latitudes. Those species which do not disperse as readily as the others will be dependent upon these changes in symbiosis or may well become locally or globally extinct [78]. It is important to understand that most species that have ever existed on Earth are now extinct. Perhaps those species of corals which are able to adapt to changing temperature regimes on the earth will represent the new members of coral communities, particularly in the hyper-tropical zone, in the future.

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References

- [1] A. Huppert, L. Stone, Chaos in the Pacific's coral bleaching cycle, *Am. Nat.* 152 (1998) 447–458.
- [2] O. Hoegh-Guldberg, Climate change, coral bleaching, and the future of the world's coral reefs, *Mar. Freshw. Res.* 50 (1999) 839–866.

- [3] Ove Hoegh-Guldberg, Low coral cover in a high-CO₂ world, 2005.
- [4] A.C. Baker, Flexibility and specificity in Coral–Algal symbiosis: Diversity, ecology, and biogeography of *Symbiodinium*, *Annu. Rev. Ecol. Evol. Syst.* 34 (2003) 661–689. <http://www.jstor.org/stable/30033790>.
- [5] C.E. Birkeland, *The Life and Death of Coral Reefs*, Springer, New York, 1997, p. 536. HS98.
- [6] P.W. Glynn, Extensive bleaching and death of reef corals on the Pacific coast of Panama, *Environ. Conserv.* 10 (1983) 149–154.
- [7] P.W. Glynn, Widespread coral mortality and the 1982–1983 El Niño warming event, *Environ. Conserv.* 11 (1984) 133–146.
- [8] UNESCO World Heritage Committee Climate Change Working Group, 2006. Potential climate change impacts on the Great Barrier Reef (Australia). Annex 4. UNESCO, Paris.
- [9] UNEP (United Nations Environment Programme), Migratory species and climate change: Impacts of a changing environment on wild animals UNEP/CMS Secretariat, Bonn, Germany, 2008, p. 68.
- [10] Australian Government, Climate change—potential impacts and costs, Great Barrier Reef, 2008, p. 3. Available at: <http://www.climatechange.gov.au/impacts/publications/pubs/fs-qld.pdf>.
- [11] J.W. Porter, O.W. Meier, Quantification of loss and change in Floridian reef coral populations, *Am. Zool.* 32 (1992) 625–640.
- [12] P.W. Glynn, Coral reef bleaching in the 1980s and possible connections with global warming, *Trends Ecol. Evol.* 6 (1991) 175–179.
- [13] P.W. Glynn, Coral reef bleaching: Ecological perspectives, *Coral Reefs* 12 (1993) 1–7.
- [14] K.B. Strychar, M.C. Coates, P.W. Sammarco, T.J. Piva, Bleaching as a pathogenic response in scleractinian corals, evidenced by high concentrations of apoptotic and necrotic zooxanthellae, *J. Exp. Mar. Biol. Ecol.* 304 (2004) 99–121.
- [15] K.B. Strychar, P.W. Sammarco, Exaptation in corals to high seawater temperatures: Low concentrations of apoptotic and necrotic cells in host coral tissue under bleaching conditions, *J. Exp. Mar. Biol. Ecol.* 369 (2009) 31–42.
- [16] P.W. Sammarco, K.B. Strychar, Responses to high seawater temperatures in zooxanthellae Octocorals, *PLoS One* 8 (2) (2013) e54989. <http://dx.doi.org/10.1371/journal.pone.0054989>.
- [17] M.J.H. Van Oppen, J.C. Mieog, C.A. Sanchez, K.E. Fabricius, Diversity of algal endosymbionts (zooxanthellae) in octocorals: The roles of geography and host relationships, *Mol. Ecol.* 14 (2005) 2403–2417. <http://dx.doi.org/10.1111/j.1365-294X.2005.02545.x>.
- [18] M.J.H. Van Oppen, J.M. Lough (Eds.), *Coral Bleaching: Patterns, Processes, Causes, and Consequences*, in: *Ecol. Studies*, 205, Springer, New York, 2009, p. 178.
- [19] M. Corder, Australia's Great Barrier Reef at risk of mass bleaching in coming months, scientists say. Associated Press Writer AP Worldstream, 01-31-2006, 2006. Available at: <http://www.highbeam.com/doc/1P1-118177864.html>.
- [20] O. Hoegh-Guldberg, P.J. Mumby, A.J. Hooten, R.S. Steneck, P. Greenfield, E. Gomez, C.D. Harvell, P.F. Sale, A.J. Edwards, K. Caldeira, N. Knowlton, C.M. Eakin, R. Iglesias-Prieto, N. Muthiga, R.H. Bradbury, A. Dubi, M.E. Hatziolos, Coral reefs under rapid climate change and ocean acidification, *Science* 318 (2007) 1737–1742. <http://dx.doi.org/10.1126/science.1152509>.
- [21] T.P. Hughes, A.H. Baird, D.R. Bellwood, M. Card, S.R. Connolly, C. Folke, R. Grosberg, O. Hoegh-Guldberg, J.B.C. Jackson, J. Kleypas, J.M. Lough, P. Marshall, M. Nyström, S.R. Palumbi, J.M. Pandolfi, B. Rosen, J. Roughgarden, Climate change, human impacts, and the resilience of coral reefs, *Science* 301 (2003) 929–933. <http://dx.doi.org/10.1126/science.1085046>.
- [22] J. Levinton, *Marine Biology: Function, Biodiversity, Ecology*, fourth ed., Oxford University Press, ASIN: B00HQMYXY, 2014.
- [23] E.A. Verde, L.R. McCloskey, Production, respiration, and photophysiology of the mangrove jellyfish *Cassiopea xamachana* symbiotic with zooxanthellae: effect of jellyfish size and season, *Mar. Ecol. Prog. Ser.* 168 (1998) 147–162.
- [24] D.L. Taylor, The cellular interactions of algal-invertebrate symbiosis, *Adv. Mar. Biol.* 11 (1973) 1–56.
- [25] Lindquist, Hay, Can small rare prey be chemically defended? The case for marine larvae, 1995.
- [26] R.K. Trench, R.J. Blank, *Symbiodinium microadriaticum* Freudenthal, *Symbiodinium goreauii* sp. nov., *Symbiodinium kawagutii* sp. nov. and *Symbiodinium pilosum* sp. nov.: gymnodinioid dinoflagellate symbionts of marine invertebrates, *J. Phycol.* 23 (1987) 469–481.
- [27] L. Muscatine, R.W. Green, Chloroplasts and algae as symbionts in molluscs, *Int. Rev. Cytol.* 36 (1973) 137–169.
- [28] W.K. Fitt, R.K. Trench, Spawning, development, and acquisition of zooxanthellae by *Tridacna squamosa* (Mollusca, Bivalvia), *Biol. Bull.* 161 (1881) 213–235.
- [29] S.F. Perez, C.B. Cook, W.R. Brook, The role of symbiotic dinoflagellates in the temperature-induced bleaching response of the subtropical sea anemone *Aiptasia pallida*, *J. Exp. Mar. Biol. Ecol.* 256 (2001) 1–14.
- [30] S.R. Dunn, J.C. Bythell, M.D.A. Le Tissier, et al., Programmed cell death and cell necrosis activity during hyperthermic stress-induced bleaching of the symbiotic sea anemone *Aiptasia* sp., *J. Exp. Mar. Biol. Ecol.* 272 (2002) 29–53. [http://dx.doi.org/10.1016/S0022-0981\(02\)00036-9](http://dx.doi.org/10.1016/S0022-0981(02)00036-9).
- [31] J.H. Norton, H.C. Prior, B. Baillie, et al., Atrophy of the zooxanthellal tubular system in bleached giant clams *Tridacna gigas*, *J. Invertebr. Pathol.* 66 (1995) 307–310. <http://dx.doi.org/10.1006/jipa.1995.1106>.
- [32] L. Margulis, Whittaker's five kingdoms of organisms: Minor revisions suggested by considerations of the origin of mitosis, *Evolution* 25 (1971) 242–245.
- [33] L. Margulis, R. Fester (Eds.), *Symbiosis as a Source of Evolutionary Innovation: Speciation and Morphogenesis*, Massachusetts Institute of Technology, Cambridge, MA, 1991, p. 437+.
- [34] G.D. Stanley Jr., B. Van De Schootbrugge, The evolution of the coral–algal symbiosis, in: *Coral Bleaching*, Springer, Berlin, Heidelberg, 2009, pp. 7–19.
- [35] K.B. Strychar, P.W. Sammarco, Zooxanthellate vs. azooxanthellate corals: Insights into susceptibility to disease and efficacy of immune systems, in: *Conference of the Int. Soc. Envtl. Indicators*, University of Science and Technology, Hefei, Anhui Province, China, 2010.

- [36] J.M. Daida, C.S. Grasso, S.A. Stanhope, S.J. Ross, Symbioticism and complex adaptive systems I: Implications of having symbiosis occur in nature, in: *Proc. Ann. Conf. Evolution. Program.*, Vol. 186, MIT Press, Cambridge, MA, USA, 1996.
- [37] P.L. Antonelli, P.W. Sammarco, Evolution of multiple complementary (secondary) metabolites, their synergism and their stability in colonial organisms, *Open Syst. Inf. Dyn.* 6 (1) (1999).
- [38] P.L. Antonelli, B.C. Lackey (Eds.), *The Theory of Finslerian Laplacians*, in: *Mathematics and its Applications*, vol. 459, Springer/Kluwer Series, Dordrecht, 1998.
- [39] M. Nagasawa, Segregation of a population in an environment, *J. Math. Biol.* 9 (2013) (1980) 213–235.
- [40] P.L. Antonelli, R.S. Ingarden, M. Matsumoto, *The Theory of Sprays and Finsler Spaces with Applications in Physics and Biology*, Springer/Kluwer, FTPH 58 Dordrecht, 1993.
- [41] E. Cartan, Observations sur le memoir precedent, *Math. Z.* 37 (1933) 619–622.
- [42] S.S. Chern, Sur la geometrie d'un systeme d'equations differentielles du second ordre, *Bull. Sci. Math.* 63 (1939) 206–212.
- [43] D. Kosambi, Systems of differential equations of second order, *Quart. J. Math. Oxford Ser.* 6 (1935) 1–12.
- [44] S.F. Rutz, Differential Geometric Modelling in Trophodynamics, in: *Proceedings of the First Brazilian Symposium on Mathematical and Computational Biology*, Rubem Mondaini (ed), p. 320, ISBN 85-87922-26-2, E-papers, Rio de Janeiro, 2001.
- [45] S.F. Rutz, R. Portugal, FINSLER: A Computer Algebra Package for Finsler Geometry, *Nonlinear Anal.* 47 (2001) 6121–6134. and in [54], 2nd volume.
- [46] L. Margulis, *Symbiosis in Cell Evolution*, W.H. Freeman and Co., New York, 1981, p. 452.
- [47] L. Margulis, in: L. Margulis, R. Fester (Eds.), *Symbiosis as a Source of Evolutionary Innovation: Speciation and Morphogenesis*, Massachusetts Institute of Technology, Cambridge, MA, 1991, p. 437+.
- [48] P.L. Antonelli, R.H. Bradbury, Volterra-Hamilton Models in the Ecology and Evolution of Colonial Organisms, in: *Series in Mathematical Biology and Medicine*, World Scientific, Singapore, 1996.
- [49] G.E. Hutchinson, A note of the theory of competition between social species, *Ecology* 28 (1947) 319–321.
- [50] P.L. Antonelli, S.F. Rutz, C.E. Hirakawa, The mathematical theory of endosymbiosis I, *Nonlinear Analysis RWA* 12 (2011) 3238–3251.
- [51] P.L. Antonelli, S.F. Rutz, K.T. Fonseca, The mathematical theory of symbiogenesis II: Models of the fungal fusion hypothesis, *Nonlinear Analysis RWA* (2012) <http://dx.doi.org/10.1016/j.nonrwa.2012.01.005>.
- [52] P.L. Antonelli, L. Bevilacqua, S.F. Rutz, Theories and models in symbiogenesis, *Nonlinear Analysis RWA* 4 (2003) 743–753.
- [53] G.D. Stanley Jr., P.K. Swart, Evolution of the coral-zooxanthellae symbiosis during the Triassic: A geochemical approach—Paleobiology, 1995.
- [54] P.L. Antonelli (Ed.), *Handbook of Finsler Geometry*, Vol. 2, Springer/Kluwer, Dordrecht, 2003.
- [55] P.L. Antonelli, T.J. Zastawniak, *Fundamentals of Finslerian Diffusion with Applications*, Springer/Kluwer, Dordrecht, FTPH 101, 1999.
- [56] P.L. Antonelli, S.F. Rutz, K.B. Strychar, P.W. Sammarco, A coral bleaching model, *Nonlinear Analysis RWA* (2013) <http://dx.doi.org/10.1016/j.nonrwa.2013.09.006>.
- [57] <http://www.maplesoft.com/>.
- [58] E. Nelson, *Quantum Fluctuations*, Princeton U. Press, 1985.
- [59] R. Rowan, Review—Diversity and ecology of zooxanthellae on coral reefs, *J. Phycol.* 34 (1998) 407–417. <http://dx.doi.org/10.1046/j.1529-8817.1998.340407.x>.
- [60] J. Needham, *Chemical Embriology*, Cambridge U. Press, 1931, 3 volumes.
- [61] S.J. Gould, *Ontogeny and Phylogeny*, Belknap Press, Harvard University Press, Cambridge, Massachusetts, USA, 1977.
- [62] P.W. Sammarco, K.B. Strychar, Effects of climate change on coral reefs: Adaptation/exaptation in corals, evolution in zooxanthellae, and biogeographic shifts, *Environ. Bioindic.* 4 (2009) 9–45.
- [63] J.E.N. Veron, *Corals in Space and Time: The Biogeography and Evolution of the Scleractinia*, Univ. New South Wales Press, Sydney, 1995, p. 321.
- [64] J. Huxley, *Problems of Relative Growth*, Dover Press, 1972.
- [65] K.B. Strychar, P.W. Sammarco, Memory in corals allows faster adaptation to global warming and changing environmental conditions, in: 3rd Int. Conference on Oceanography. June 22–24, Philadelphia, USA, 2015.
- [66] K.B. Strychar, P.W. Sammarco, Does climate change drive immunological memory in lower marine invertebrates, in: 2nd Int. Conference on Oceanography. July 21–23, Las Vegas, USA, 2014.
- [67] A.E. Douglas, Coral bleaching—how and why? *Mar. Pollut. Bull.* 46 (2003) 385–392.
- [68] J.L. Harper, *The Population Biology of Plants*, Academic Press, New York, 1977.
- [69] M.C. Brundrett, Coevolution of roots and mycorrhizas of land plants, *New Phytol. Tansley Rev.* 154 (134) (2002) 275–304.
- [70] T. Fukatsu, H. Ishikawa, Phylogenetic position of yeast-like symbiont of *Hamiltonaphis styraci* (Homoptera, Aphididae) based on 18S-rDNA sequence, *Insect Biochem. Mol. Biol.* 26 (1996) 383–388.
- [71] J.F. Saldarriaga, F.J.R. Taylor, P.J. Keeling, T. Cavalier-Smith, Dinoflagellate nuclear SSU rRNA phylogeny suggests multiple plastid losses and replacements, *J. Mol. Evol.* 53 (2001) 204–213.
- [72] C.A. Belda-Baillie, B.K. Ballie, T. Maruyama, Specificity of a model cnidarian-dinoflagellate symbiosis, *Biol. Bull.* 202 (2002) 74–85.
- [73] A.W.D. Larkum, The Cyanobacteria of coral reefs. Marine cyanobacteria, in: L., Charpy, A.W.D., Larkum, *Bull. de Oceanographique Monaco, Numeral Special 19*, Monaco Musee Oceanographique, 1999, pp. 149–167.
- [74] J. Dolan, Myxotrophy in ciliates: a review of *Chlorella* symbiosis and chloroplast retention, *Mar. Microb. Food Webs* 6 (1992) 115–132.
- [75] M.D. Johnson, Acquired phototrophy in ciliates: a review of cellular interactions and structural adaptations, *J. Eukaryot. Microbiol.* 58 (2011) 185–195. <http://dx.doi.org/10.1111/j.1550-7408.2011.00545.x>.

- [76] A. Ritchie, T. Lewis, Bacterial composition of surface associated microbes found in three Hawaiian coral species: *Porites compressa*, *Porites lobata*, and *Montipora capitata*, *J. Young Investig.* 17 (5) (2005).
- [77] G. Christa, K. Händeler, P. Kück, M. Vleugels, J. Franken, D. Karameinski, Phylogenetic evidence for multiple independent origins of functional kleptoplasty in *Sacoglossa* (Heterobranchia, Gastropoda), *Org. Divers. Evol.* 15 (2015) 23–36.
- [78] P.W. Sammarco, K.B. Strychar, Ecological and evolutionary considerations regarding corals in a rapidly changing environment, in: S. Goffredo, Z. Dubinsky (Eds.), *The Cnidaria Past, Present, and Future: The world of Medusa and her sisters*, Springer, New York, 2016, in press.